

Detailed Use Case Analysis: Healthcare Management System (HCMS)

This analysis is based on the provided Use Case Diagram for the Healthcare Management System, expanding upon the high-level functional requirements defined in the initial case study. It details the specific actions (Use Cases) performed by each system **Actor**.

1. Actor Roles and Responsibilities

The system identifies four primary actors, each with distinct interaction profiles:

Actor	Primary Role in the System	Key Functions
Patient	Consumer of healthcare services; manages their appointments and personal health information.	Booking, registration, viewing records.
Nurse	Clinical support role; responsible for patient intake and record maintenance.	Registration, updating records.
Doctor	Primary care provider; responsible for treatment, scheduling, and prescribing.	Treatment, scheduling, record maintenance.
Admin	System manager; responsible for user accounts, finance, and system oversight.	Billing, reporting, staff management.

2. Core Use Case Categories

The system's functionality is grouped into four logical categories:

2.1. System Access and Security

This use case is fundamental for all actors to securely enter the system.

- **Use Case: Register/Login**
 - **Actors:** Patient, Nurse, Doctor, Admin
 - **Description:** Allows a user to create a new account (Register) or authenticate their identity (Login) to access system features corresponding to their role.

2.2. Patient Interaction and Scheduling

These functions directly support the patient experience and care coordination.

- **Use Case: Book Appointment**
 - **Actor:** Patient
 - **Description:** Allows the patient to view available slots for doctors or services and schedule a new appointment.
- **Use Case: Manage Schedule**
 - **Actor:** Doctor
 - **Description:** Allows the Doctor to view, modify, accept, or decline scheduled appointments, and manage their availability calendar.

2.3. Clinical and Records Management

These are the core clinical functions ensuring accurate documentation and treatment.

- **Use Case: View Medical History**
 - **Actor:** Patient
 - **Description:** Allows the Patient to securely review their past visits, diagnoses, treatment plans, and test results.
- **Use Case: Update Patient Records**
 - **Actors:** Nurse, Doctor
 - **Description:** Allows clinical staff to input new vital signs, consultation notes, lab results, and patient progress information into the electronic medical record (EMR).
- **Use Case: Prescribe Medication**
 - **Actor:** Doctor
 - **Description:** Allows the Doctor to electronically generate, authorize, and submit prescriptions, including dosage, frequency, and duration, often linked to an external pharmacy system.

2.4. Administrative and Financial Management

These functions ensure the smooth operation and financial health of the facility.

- **Use Case: Manage Staff Accounts**
 - **Actor:** Admin

- **Description:** Allows the Administrator to create, modify, suspend, or terminate accounts for all Staff, Nurses, and Doctors, ensuring proper access control. (Corresponds to "Create Staff Account" from the DFD).
- **Use Case: Process Billing**
 - **Actor:** Admin
 - **Description:** Allows the Administrator to generate patient invoices, apply insurance claims, record payments, and manage all financial transactions related to patient care. (Corresponds to "Payment" and "Discharge Bill" from the DFD).
- **Use Case: Generate Reports**
 - **Actor:** Admin
 - **Description:** Allows the Administrator to extract, filter, and compile reports on financial, operational, and clinical data for strategic planning and auditing. (Corresponds to "Report" and "Hospital Reports" from the DFD).

3. Linkage to Previous Analysis

The Use Case Diagram confirms and elaborates on the system's external interactions identified in the Context DFD:

DFD Data Flow	Corresponding Use Case(s)	Actor
Create Staff Account	Manage Staff Accounts	Admin
Add Master Records	Update Patient Records (Master data is integrated into clinical records)	Nurse, Doctor
Payment	Process Billing	Admin (Receives/processes)
Discharge Bill	Process Billing	Admin (Generates)
Report / Hospital Reports	Generate Reports	Admin

(New Function)	Book Appointment, Prescribe Medication, Manage Schedule, View Medical History	Patient, Doctor
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